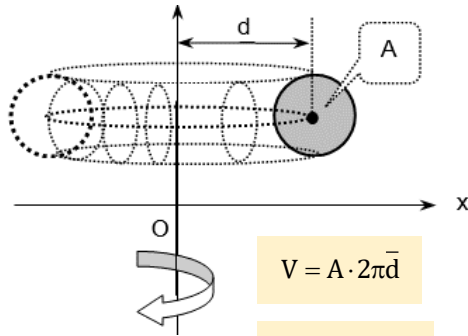




Integral Calculus Applications

VOLUME continuation...



$$V = A \cdot 2\pi d$$

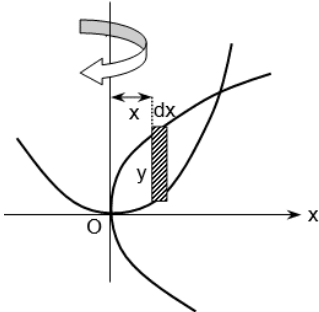
$$V = \int A \cdot 2\pi d$$

where:

d = distance from centroid to axis of rotation

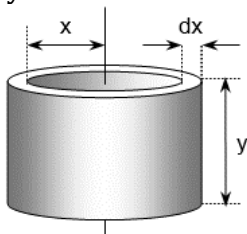
If the area is not given or cannot be computed easily, the following will be used in finding the volume of the solid:

A. Using a vertical differential strip (Shell Method):



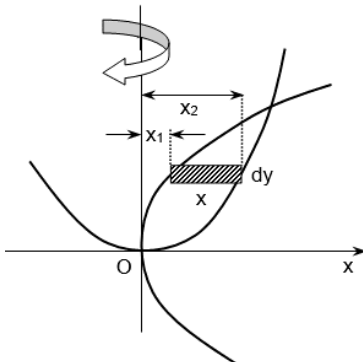
The figure formed by rotating the vertical strip about the vertical axis is a **hollow cylindrical shell**.

Excel Review Center

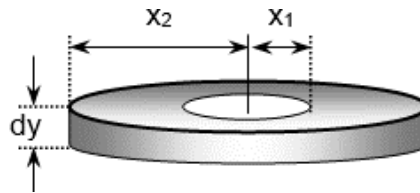


$$V = \int 2\pi y x dx$$

B. Using a horizontal differential strip (Ring Method):



The figure formed by rotating the horizontal strip about the vertical axis is a **circular ring or washer**



$$V = \int \pi(x_2^2 - x_1^2) dy$$

Excel Review Center

6. Work

Work is defined as the force multiplied by the distance traversed. If the force is constant throughout the distance, then work is simply the product of the applied force and the distance traveled. However, if the applied force varies with the distance, then the best solution is to apply integration.

$$\text{Work} = \int F \cdot dx$$

where:

F = force expressed in terms of x

Excel Review Center

Hooke's Law

Hooke's Law, named after the English mathematician and scientist **Robert Hooke** (1635 – 1703) is stated as follows:

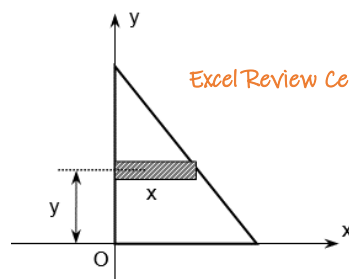
"Within elastic limit, the force required to stretch (or compress) a spring is proportional to the change in length."

For spring:

$$F = kx$$

where: k = spring constant

7. Moment of Inertia



Excel Review Center

$$I_x = \int dA \cdot y^2$$

$$I_y = \int dA \cdot \left(\frac{x}{2}\right)^2$$

Multiple Integrals

An integral in which the integrand is integrated twice is a **double integral**. Example:

$$\iint f(x) dx^2$$

If integrated thrice, the integral is called **triple integral**. Example:

$$\iiint f(x) dx^3$$

An integral in which the integrand is integrated to n times is **n-fold iterated integral**. Example:

$$\int \dots \int f(x) dx^n$$

Functions of more than one variable may be integrated with respect to one variable at a time while the other variables are held constant, reversing the process of partial differentiation.

Excel Review Center